UHG Provider Intelligence Platform — Enterprise PPT Content Document

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Slide 1 — Business Context & Problem Statement

Current Challenges

- Manual provider comparison across competitor carrier networks
- Inconsistent and incomplete provider datasets
- Missing NPIs and duplicate provider records
- Time-consuming disruption analysis workflows
- Limited scalability across carriers and product lines
- High dependency on manual analyst effort
- Risk of inaccurate provider continuity analysis
- Operational delays impacting competitive bids

Business Problem Statement

UHG faces persistent challenges in accurately identifying and mapping provider differences between competitor carrier networks and UHG networks. Current manual workflows are inefficient, difficult to scale, and prone to inconsistencies, leading to operational delays and potential loss of competitive opportunities.

Proposed Solution

- Automate provider data ingestion, normalization, and validation workflows
- Enable high-confidence provider matching across competitor and UHG networks
- Support deterministic, probabilistic, and AI-assisted provider resolution
- Enrich incomplete provider records using external healthcare registries
- Generate automated disruption and provider continuity analysis reports
- Provide explainable and confidence-driven matching workflows
- Enable human validation for low-confidence or ambiguous provider matches
- Reduce manual operational effort and improve workflow scalability

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Expected Business Impact

Area	Expected Impact
Manual Analysis Effort	70–85% Reduction
Provider Matching Accuracy	≥98% Target
Operational Efficiency	Improved
Carrier Scalability	Multi-carrier support
Sales Support	Faster disruption analysis
Governance	Explainable workflows
Enterprise Readiness	Scalable architecture

Slide 2 — Implementation Foundation & Dependency Planning

Customer Environment Readiness

Required from Customer

- Development, UAT, and Production environments
- Cloud/on-prem deployment decision
- Network and VPN connectivity
- Security and access approvals
- Authentication/SSO integration requirements
- Infrastructure provisioning approach

Data & Business Dependencies

Required Inputs

- Sample provider/carrier datasets
- Historical datasets for validation
- Existing provider matching rules
- Business formulas and confidence thresholds
- Existing provider master/reference datasets
- Disruption analysis expectations

External Integration Dependencies

Required Access

- NPPES access
- CMS/PECOS access
- Internal provider systems (if applicable)
- API/network whitelisting
- External enrichment approval

Governance & Operational Dependencies

Required Alignment

- HITL reviewer identification
- Match validation process
- UAT ownership
- Auditability expectations
- Explainability requirements
- Production support model

Slide 3 — Enterprise Workflow & Capability Architecture

Core Workflow Sequence

Multi-File Ingestion & Schema Normalization ↓ Deterministic Cleaning & Standardization ↓ Tiered Deduplication ↓ Deterministic Provider Matching ↓ External Validation & Enrichment ↓ Selective AI-Assisted Resolution ↓ Confidence Threshold Validation ↓ Human-in-the-Loop Review ↓ Disruption Analysis & Explainable Reporting ↓ GIS Visualization & Geographic Intelligence

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Workflow Objectives

Workflow Stage	Objective
Multi-File Ingestion	Process heterogeneous provider datasets from multiple carriers
Cleaning & Standardization	Improve provider data quality and consistency
Deduplication	Remove duplicate provider entities

Workflow Stage	Objective
Deterministic Matching	Enable high-confidence provider resolution
Enrichment	Validate and enrich providers using external sources
AI-Assisted Resolution	Resolve residual ambiguous provider cases
HITL Validation	Govern low-confidence provider decisions
Disruption Analysis	Generate provider continuity intelligence
GIS Intelligence	Provide geographic visibility and regional insights

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Slide 4 — Multi-File Ingestion & Schema Normalization

Objective

Enable ingestion and processing of heterogeneous provider datasets from multiple carrier sources while preserving source lineage and preparing standardized datasets for downstream workflows.

Core Operations

- Multi-file CSV/XLSX ingestion
- Immutable raw file storage
- Schema detection and normalization
- Source-to-canonical column mapping
- File and row-level lineage tracking
- Ingestion metadata registration
- Validation summary generation

Customer Inputs Required

- Sample carrier/provider datasets
- UHG reference datasets
- Expected file formats and schema variations
- Mandatory provider fields
- File delivery mechanism (SFTP/API/manual upload)

Dependencies & Technical Requirements

- Blob/object storage for raw file retention
- Metadata database for lineage tracking
- Schema mapping configuration
- File validation framework
- Network and file transfer setup
- Historical data retention requirements

Validations Performed

- File format validation
- Mandatory column validation
- Schema mismatch checks
- Corrupted file detection
- Duplicate upload validation
- Row-level parsing validation

Outputs Generated

- Immutable raw file repository
- Normalized canonical datasets
- File-level metadata
- Row-level lineage mapping
- Validation reports
- Processing-ready datasets

Slide 5 — Deterministic Cleaning, Deduplication & Provider Matching

Cleaning & Standardization

Core Operations

- Provider name normalization
- Address standardization
- Taxonomy and specialty standardization
- NPI, ZIP, and TIN validation
- Invalid character cleanup
- Deterministic rule-based cleansing

Outputs

- Standardized provider datasets
- Cleansed provider records
- Data quality metrics
- Transformation audit logs

Tiered Deduplication

Core Operations

- Deterministic exact deduplication
- Rule-based duplicate detection
- Probabilistic/fuzzy deduplication
- AI-assisted residual deduplication
- Duplicate clustering and consolidation
- Confidence scoring

Outputs

- Deduplicated provider datasets
- Duplicate mapping relationships
- Residual ambiguous duplicate cases
- Deduplication audit logs

Deterministic Provider Matching

Core Operations

- Exact NPI matching
- Tax ID and provider identifier matching
- Provider name matching
- Address and ZIP-based matching
- Taxonomy validation
- Rule-based provider resolution
- Match confidence classification

Outputs

- High-confidence matched providers
- Unmatched provider records
- Match confidence scores
- Match evidence and traceability

Slide 6 — External Validation, AI-Assisted Resolution & HITL

External Validation & Enrichment

Core Operations

- NPPES provider validation
- CMS/PECOS verification
- Internal provider reference enrichment
- Missing attribute augmentation
- Source attribution and evidence tagging
- Append-only enrichment processing

Outputs

- Enriched provider records
- Validation evidence metadata
- Source-attributed enrichment fields
- Confidence-enhanced provider profiles

Selective AI-Assisted Resolution

Core Operations

- Residual ambiguous provider analysis
- Semantic similarity evaluation
- Context-aware provider comparison
- AI-based candidate ranking
- Confidence score generation
- Residual unresolved case identification

Outputs

- AI-assisted provider recommendations
- Confidence-scored candidate rankings
- Explainability metadata
- HITL review candidates

Confidence Threshold Validation & HITL

Core Operations

- Confidence threshold evaluation
- Auto-approval for high-confidence matches
- Low-confidence case routing
- Human reviewer validation workflows
- Override and escalation handling
- Reviewer audit logging

Outputs

- Approved provider matches
- Reviewed and validated records
- Escalated unresolved cases
- Reviewer audit history

Slide 7 — Disruption Analysis, GIS Intelligence & Governance

Disruption Analysis & Explainable Reporting

Core Operations

- Competitor vs UHG network comparison
- Provider gap and disruption analysis
- Continuity impact assessment
- Match success and failure analysis
- Explainable provider-level reporting
- Carrier-wise comparative reporting

Outputs

- Disruption analysis reports
- Provider continuity insights
- Carrier comparison reports
- Executive operational dashboards

GIS Visualization & Geographic Intelligence

Core Operations

- Geographic provider mapping
- Regional disruption visualization
- Provider density analysis
- Heatmap generation
- Carrier-wise geographic comparison
- ZIP/state-level continuity analysis

Outputs

- Geographic disruption heatmaps
- Regional provider coverage analysis
- Carrier-wise geographic insights
- GIS dashboards and visual analytics

Platform Governance & Auditability

Core Operations

- End-to-end workflow audit logging
- File and row-level lineage tracking
- Historical run retention
- Change and evidence tracking
- Reviewer action auditing
- Workflow traceability and reporting

Outputs

- End-to-end audit trails
- Historical workflow records
- Explainable provider matching evidence
- Governance and compliance logs

Slide 8 — Production Architecture

Architecture Guidance

Use the uploaded architecture image as the primary visual.

Architecture Should Include

- User Layer
- Frontend/UI Layer
- Workflow Orchestration Layer
- Agent Framework
- Provider Matching Engine
- Azure OpenAI Integration
- External Enrichment Integrations
- Data Storage Layer
- Reporting & GIS Layer
- Monitoring & Audit Layer

Technology Stack

Layer	Technologies
Frontend	React / Streamlit
Backend APIs	Python FastAPI
AI & Embeddings	Azure OpenAI
Vector Search	FAISS
Workflow Orchestration	Agent-based orchestration framework
Data Processing	Pandas / Python
Storage	Blob/Object Storage
Reporting & GIS	Azure Maps / GIS Frameworks
Deployment	Azure VM / Containers / Kubernetes
Monitoring	Logging & Monitoring Frameworks

Slide 9 — Delivery Roadmap & Implementation Timeline

Phase	Priority	Estimated Duration	Key Deliverables
Environment Readiness & Infrastructure Setup	Critical	Weeks 1–3	Infrastructure foundation
Dataset Onboarding & Business Rule Alignment	Critical	Weeks 4–5	Business-aligned rule framework

Phase	Priority	Estimated Duration	Key Deliverables
Multi-File Ingestion & Schema Normalization	High	Weeks 6–7	Standardized ingestion workflows
Deterministic Cleaning & Standardization	High	Weeks 8–9	Cleansed provider datasets
Tiered Deduplication Framework	High	Weeks 10–11	Deduplicated provider entities
Deterministic Provider Matching Engine	Critical	Weeks 12–14	Provider matching workflows
External Validation & Enrichment	High	Weeks 15–16	Enrichment workflows
AI-Assisted Resolution & HITL Enablement	Medium	Weeks 17–18	Governed AI workflows
Disruption Analysis & Explainable Reporting	High	Weeks 19–20	Reporting framework
GIS Visualization & Geographic Intelligence	Medium	Weeks 21–22	GIS analytics and dashboards
System Integration Testing & Performance Validation	Critical	Weeks 23–24	Production-validated platform
UAT, Production Rollout & Hypercare	Critical	Weeks 25–26	Production-ready platform

Slide 10 — Expected Outcomes Across Delivery Timeline

Timeline	Focus Area	Key Outcomes
Month 1	Environment & Foundation Setup	Infrastructure and connectivity readiness
Month 2	Ingestion, Cleaning & Standardization	Operational data processing workflows
Month 3	Deduplication & Deterministic Matching	High-confidence provider matching
Month 4	External Validation & AI Workflows	AI-assisted provider resolution
Month 5	Reporting & GIS Intelligence	Disruption analysis and GIS dashboards
Month 6	UAT, Production Rollout & Stabilization	Production-ready provider intelligence platform

Overall Expected Outcomes

- 70–85% reduction in manual disruption analysis effort
- ≥98% targeted provider matching accuracy
- Multi-carrier operational scalability
- Explainable and confidence-driven workflows
- Faster provider continuity analysis
- Production-ready provider intelligence platform

Slide 11 — Discussion Points & Customer Alignment Areas

Environment & Deployment Alignment

- Cloud vs On-Prem deployment strategy
- Development/UAT/Production environment availability
- Network, VPN, and firewall constraints
- Authentication and SSO integration requirements
- Infrastructure provisioning ownership

Data & Business Rule Alignment

- Sample carrier/provider dataset availability
- Historical dataset access for validation
- Provider matching business rules
- Confidence threshold expectations
- Disruption analysis and continuity logic

External Integration Alignment

- NPPES and CMS access availability
- Internal provider system integrations
- API/network whitelisting requirements
- GIS/geocoding integration expectations
- External enrichment approval workflows

Governance & Operational Alignment

- HITL reviewer identification

- Match approval and escalation workflows
- Explainability and audit expectations
- UAT ownership and sign-off process
- Production support and monitoring ownership

Implementation Readiness Discussion

- Expected delivery timelines
- Infrastructure readiness dependencies
- Validation and UAT expectations
- Scalability and performance expectations
- Security and compliance considerations